
Schema-Adaptive Action-Conditioned JEPA for Cross-Machine CNC Transfer under Partial Sensor Overlap

Ayoub Louaye Bouaziz
Université de Bretagne Occidentale
Brest, France
abouaziz@univ-brest.fr

Matthieu Joseph Paul Ostertag Tressoux
Université de Lorraine
Nancy, France
matthieu.ostertag7@etu.univ-lorraine.fr

Anton Demasles
Université de Lorraine
Nancy, France
anton.demasles2@etu.univ-lorraine.fr

Abstract

Cross-machine deployment of industrial world models requires transfer across changes in machine dynamics, sensing interfaces, sampling regimes, and control units. This study investigates a schema-adaptive action-conditioned Joint-Embedding Predictive Architecture (SAAC-JEPA) for CNC dynamics when a source machine provides 17 canonical sensor channels and a target machine shares only 10. The evaluation protocol contains group-disjoint source splits, source-only normalization, deterministic held-out self-supervised validation, explicit unit audits, and a sealed target-machine test used only after model locking. A matched five-seed comparison shows no clean-source forecasting gain from JEPA pretraining: scratch and pretrained-body models obtain RMSE 0.811 ± 0.022 and 0.813 ± 0.022 , respectively. A source-only architecture search over 20 candidates then selects a schema-consistent action-conditioned JEPA variant after stability checks across seven seeds. On the single confirmatory target-machine pass, the locked model reaches zero-shot RMSE 0.546, $R^2 = 0.012$, and NLL 0.52, improving over persistence but not over official PatchTST and iTransformer zero-shot baselines (0.503 and 0.498). A pre-lock adaptation sweep further suggests that small target support can reduce RMSE to 0.520. The evidence supports a narrow conclusion: source-domain forecasting accuracy alone is insufficient for evaluating industrial predictive representations, and cross-machine adaptation under partial sensor overlap is a distinct evaluation axis for action-conditioned world models.

1 Introduction

Industrial control systems generate multivariate sensor streams while exposing controllable inputs such as spindle speed and axis feed commands. A deployable world model should represent the evolution of the physical process, predict future states under candidate actions, and remain useful when the sensing interface changes across machines. This requirement differs from conventional in-domain time-series forecasting because the target machine can exhibit both *domain shift* and *schema shift*: dynamics, controller conventions, units, sampling rates, and sensor availability can all change simultaneously.

Joint-Embedding Predictive Architectures (JEPAs) learn by predicting representations rather than reconstructing every observed sample (Assran et al., 2023; Bardes et al., 2024). This objective is Preprint.

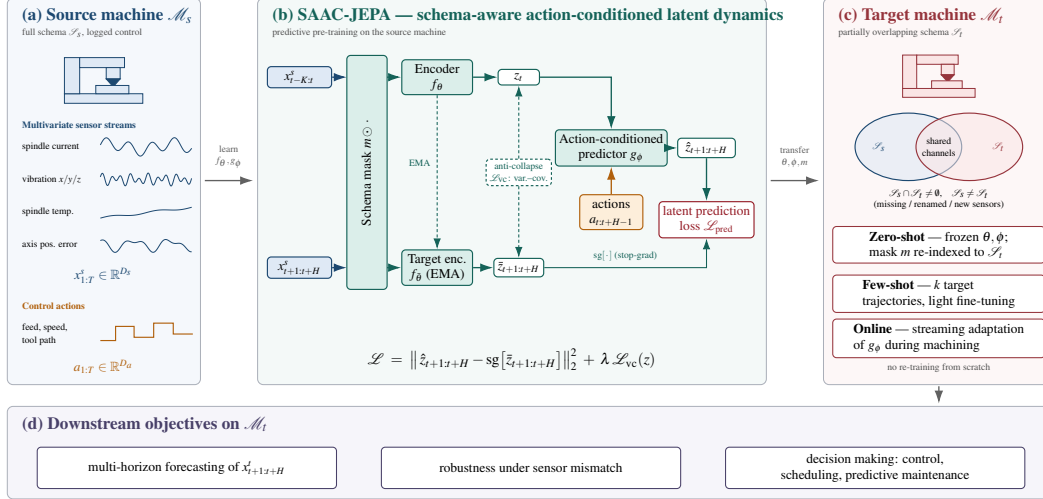


Figure 1: Overall view of SAAC-JEPA for cross-machine CNC transfer. A source CNC machine provides multivariate sensor streams and control actions, whereas the target machine exposes only a partially overlapping sensor schema. The model learns action-conditioned latent dynamics using a context encoder, an EMA target encoder, latent prediction, schema masking, and anti-collapse regularization. The learned representation is subsequently evaluated under zero-shot, few-shot, and online adaptation settings. The downstream objective is cross-machine forecasting and decision support under machine and sensor-schema shift.

potentially attractive for industrial transfer because a latent representation can emphasize predictable process dynamics while reducing dependence on machine-specific observation details. However, JEPA training is susceptible to representational collapse, and strong alternatives already exist in time-series forecasting and masked pretraining (Bardes et al., 2022; Dong et al., 2023; Nie et al., 2023; Liu et al., 2024). Action-aware industrial world modeling has also been considered previously (Yan et al., 2025). Consequently, the relevant question is not whether JEPA can be applied to industrial data, but whether schema-adaptive, action-conditioned predictive representations remain useful under cross-machine and partial-observation shift.

This study evaluates that question on a CNC source-to-target setting (Figure 1) built from two public datasets: the THWS five-axis CNC milling production dataset with multiple changeovers (Martinez et al., 2025) as the source machine, and the FH JOANNEUM CNC machining data repository with high-frequency energy measurements (Brillinger et al., 2025) as the target machine. The source machine contains 17 canonical sensor channels across 62 program sessions, whereas the target exposes a 10-channel transferable subset across 7 independent runs. The protocol was designed to prevent target leakage during model selection and includes source-only normalization, session-grouped splitting, held-out SSL validation, unit audits, collapse diagnostics, matched scratch/pretraining controls, trivial forecasting baselines, and run-level target statistics.

The contributions are as follows:

- a leakage-audited cross-machine CNC protocol combining machine shift with partial sensor overlap;
- a schema-adaptive action-conditioned JEPA formulation with latent schema consistency, EMA targets, and explicit anti-collapse regularization;
- an empirical diagnosis showing that JEPA pretraining does not improve clean source RMSE over scratch, while representation stability is sensitive to latent-loss weighting, variance–covariance regularization, and EMA dynamics;
- a pre-lock diagnostic adaptation sweep suggesting that limited target support improves cross-machine forecasting relative to zero-shot transfer; it is retained as diagnostic rather than confirmatory evidence.

2 Related Work

Joint-embedding predictive learning. I-JEPA introduced representation prediction as a non-generative self-supervised objective for images (Assran et al., 2023), later extended to video (Bardes et al., 2024); VICReg regularizes representation variance and covariance to avoid collapse (Bardes

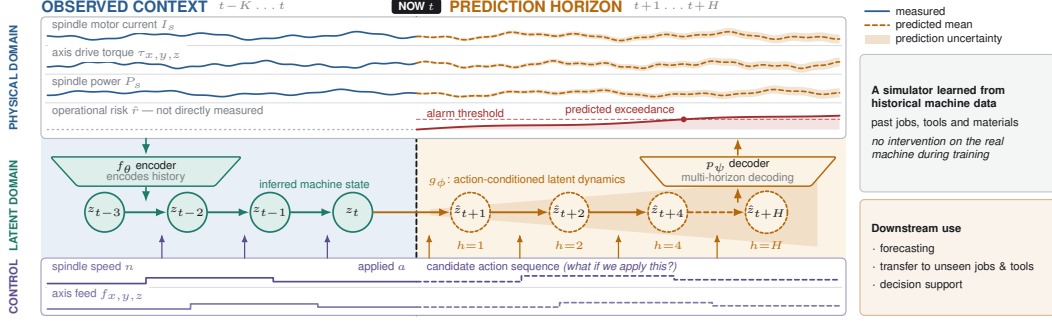


Figure 2: A CNC world model in practice. Recent measurements are encoded into a latent state, candidate future actions condition multi-horizon latent predictions, and a physical head outputs process variables with uncertainty. Conceptual schematic distinguishing passive forecasting from action-conditioned simulation of candidate futures; not a closed-loop deployment result.

et al., 2022). The CNC experiments below show that anti-collapse control is operationally necessary rather than auxiliary.

Time-series forecasting and masked pretraining. SimMTM reconstructs masked series from complementary masked neighbors (Dong et al., 2023); PatchTST uses channel-independent patch tokens (Nie et al., 2023); iTransformer inverts the tokenization axis to model multivariate dependencies (Liu et al., 2024); RevIN counters distribution shift by instance normalization (Kim et al., 2022). These methods motivate a protocol in which JEPA is not assumed to dominate source-domain forecasting and must justify its value under transfer and schema mismatch.

World models and industrial adaptation. Latent dynamics models support multi-step prediction and planning (Hafner et al., 2019), and actionable world models have been studied for industrial process control (Yan et al., 2025). The present setting adds cross-machine transfer, partial sensor overlap, explicit action alignment, source-only normalization, and post-training adaptation on the target machine.

3 Problem Formulation

Let $\mathcal{M}_s$ and $\mathcal{M}_t$ denote the source and target CNC machines, with sensor schemas $\mathcal{S}_s$ and $\mathcal{S}_t$. The source observation at time t is $x_t^s \in \mathbb{R}^{D_s}$, and the target exposes only a partially overlapping set of channels, $\mathcal{S}_s \cap \mathcal{S}_t \neq \emptyset$ and $\mathcal{S}_s \neq \mathcal{S}_t$. A binary mask $m_t \in \{0, 1\}^{D_s}$ indicates the channels available to the encoder after re-indexing to the canonical source schema. The control vector a_t contains the commanded spindle and axis-feed variables applied over $(t, t+1]$, so x_{t+1} depends on a_t in the usual world-model convention.

Given a context of length K and a future action sequence of length H , the predictive objective is

$$p_{\Theta}(x_{t+1:t+H} \mid x_{t-K:t}, m_{t-K:t}, a_{t:t+H-1}). \quad (1)$$

The forecasting metric measures numerical prediction error. The world-model requirement is stronger: predictions should also respond meaningfully to changes in future actions. This distinction is essential in highly autocorrelated industrial streams, where a model can obtain low short-horizon error while effectively ignoring control inputs. Figure 2 illustrates this operational interpretation.

4 Method

4.1 Schema-adaptive context encoding

The context encoder f_{θ} receives observed sensor values, the sensor-presence mask, and past actions:

$$z_t^c = f_{\theta}(x_{t-K:t}, m_{t-K:t}, a_{t-K:t}). \quad (2)$$

Schema masking generates multiple partial views of the same source trajectory. Missing channels are represented structurally rather than replaced before normalization, so the model receives an explicit indication of which measurements are available.

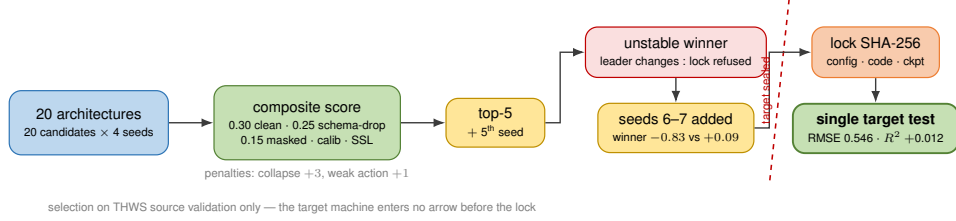


Figure 3: Leakage-free model-selection pipeline. Twenty pre-defined SAAC-JEPA candidates are ranked on THWS source validation only, first with four seeds and then with a fifth seed for the top-5. When the provisional winner changes after confirmation, locking is refused and extra seeds are added to the leading candidates. In the present study, the per-horizon VICReg variant led the discovery stage, but the schema-consistency variant with action recovery remained superior after the full stability check and was frozen before a single confirmatory evaluation on the JOANNEUM target machine.

4.2 Action-conditioned latent prediction

A predictor g_ϕ receives the context representation and candidate future actions:

$$\hat{z}_{t+1:t+H} = g_\phi(z_t^c, a_{t:t+H-1}). \quad (3)$$

In practice the context spans $K = 32$ steps and prediction is made at a fixed set of horizons $\mathcal{H} = \{1, 2, 4, 8, 16\}$, $H = 16$, by direct multi-horizon prediction rather than autoregressive rollout. Windows are formed on the 1 Hz grid (Table 1), so the context covers 32 s and the longest horizon 16 s. For each $h \in \mathcal{H}$ the predictor receives the mean of $a_{t:t+h-1}$, i.e. the mean command applied over $(t, t+h]$, so the action conditioning captures the level of the future command rather than its temporal profile. A target encoder $f_{\bar{\theta}}$ embeds the future observation window into $\bar{z}_{t+1:t+H}$. The target parameters are updated by exponential moving average,

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta, \quad (4)$$

and gradients are stopped through the target branch.

The SSL objective is

$$\mathcal{L}_{\text{SSL}} = \lambda_{\text{lat}} \mathcal{L}_{\text{lat}} + \lambda_{\text{vic}} \mathcal{L}_{\text{VICReg}} + \lambda_{\text{sch}} \mathcal{L}_{\text{schema}} + \lambda_{\text{act}} \mathcal{L}_{\text{act}}, \quad (5)$$

where $\mathcal{L}_{\text{lat}}$ aligns predicted and target latent trajectories, $\mathcal{L}_{\text{VICReg}}$ prevents low-variance or redundant embeddings, and $\mathcal{L}_{\text{schema}}$ encourages two partial sensor views of the same trajectory to predict compatible future latents. Schema consistency is imposed on latent dynamics rather than on an untrained physical output head. $\mathcal{L}_{\text{act}}$ is an action-recovery loss: a small head predicts the mean of $a_{t:t+h-1}$ from the pair $(z_t^c, \bar{z}_{t+h})$, so that the latent transition must retain information about the applied action.

4.3 Physical forecasting head and adaptation

After pretraining, a fresh probabilistic head maps latent forecasts to channel-wise means and log variances. The primary fine-tuning protocol uses a fresh head because an output head exposed during pure SSL has no supervised physical target. Three matched controls are retained: full training from scratch, pretrained body with a fresh head, and complete checkpoint transfer.

Target adaptation updates a restricted subset of parameters using target support windows. Zero-shot transfer uses the source-trained model without target updates. Few-shot transfer takes the leading fraction of each target run as support, updates the predictor for 50 steps at learning rate 10^{-4} , and evaluates on the remaining, later windows of the same runs, so support and query windows are disjoint. Online adaptation is reserved for causal prequential evaluation and is not used in the results reported in this study.

5 Experimental Protocol

5.1 Data and audits

The source machine is the Spinner U5-620 five-axis machining center of the THWS production dataset with multiple changeovers (Martinez et al., 2025); the target machine is provided by the FH

Table 1: Audited cross-machine data protocol.

	THWS source	JOANNEUM target
Raw rows	29,785	1,503,915
Independent units	62 sessions	7 runs / 13 sessions
Working rate	1 Hz	1 Hz (from 2 ms cycles)
Canonical/shared sensors	17	10 shared
Actions	spindle + commanded X/Y/Z feeds	
Normalization	source-train only	source normalizer

JOANNEUM CNC machining data repository (Brillinger et al., 2025). Both datasets are public under CC BY 4.0. The merged table (Table 1) contains 1,533,700 rows and 25 columns. THWS contributes 29,785 samples at 1 Hz, segmented into 62 NC-program sessions after excluding jog periods and splitting at temporal gaps longer than 5 s. JOANNEUM contains 1,503,915 raw controller-cycle records across 7 runs and 13 sessions. The source provides 17 canonical sensor channels, whereas 10 current/torque/power channels are shared with the target. Four mapped action variables represent spindle speed and the commanded X/Y/Z feeds.

Source sessions are divided into 42 training, 11 validation, and 9 test sessions with group-disjoint splitting. The resulting source window counts are 18,267 / 3,423 / 5,189, with zero window overlap across partitions. All normalizers are fitted on source training data only. A unit audit detected three mismatches that were corrected deterministically before modeling: JOANNEUM power channels were stored in watts while THWS used kilowatts (factor 10^{-3}); commanded axis feeds were recorded in mm/s on one machine and mm/min on the other (factor 60); and spindle speed was logged in degrees per second on one machine and rpm on the other (factor 6, verified from position derivatives). The last two matter because the action channels condition the predictor. Target statistics are not used to fit source normalizers. Validation masking is deterministic, DataLoader workers are seeded, checkpoint loading reports missing and unexpected keys, and regression tests cover split integrity, masking determinism, EMA and stop-gradient behavior, and metric aggregation. All experiments ran on one NVIDIA DGX Spark (one GB10 GPU), about six GPU-days in total.

5.2 Model selection and baselines

The source-only architecture search defines 20 candidates prior to execution. The search spans masking strategies, action-injection mechanisms, VICReg placement, latent-loss weight, horizon aggregation, schema consistency, action recovery, EMA dynamics, model capacity, normalization, and head initialization. Each candidate is first run with four seeds. Ranking uses a composite source-validation score: each component is z-scored within the candidate pool and combined as $0.30 \text{ clean RMSE} + 0.25 \text{ schema-drop RMSE} + 0.15 \text{ masked RMSE} + 0.10 \text{ long-horizon RMSE} + 0.10 \text{ calibration penalty} + 0.10 \text{ SSL validation loss}$, plus penalties of 3.0 per collapsed seed and 1.0 for weak action usage (lower is better). The top-5 candidates receive a fifth seed. If the provisional winner changes under confirmation, model locking is refused and additional seeds are added to the tied leaders. This occurred here: the per-horizon VICReg variant led the discovery stage, but the schema-consistency variant with action recovery overtook it after confirmation and remained superior after seven valid seeds, after which the configuration, code, normalizer, and checkpoint were frozen before a single confirmatory pass on the JOANNEUM target machine.

Baselines include supervised MLP, GRU, LSTM, TCN, DLinear, Transformer, TSMixer, and RSSM models, together with trivial persistence and linear-drift references. For the target comparison, the official PatchTST and iTransformer repositories are trained once on the same source-train windows (single seed, repository defaults, RevIN enabled) and evaluated on the same sealed target windows; they receive the 17 sensors plus the 4 action channels as input, and channels absent on the target are zero-imputed in normalized space (Appendix E). Six in-house reconstruction-pretraining baselines (MAE-, SimMTM-, PatchFormer-, EMIT-, MMR- and LoMaR-style) were implemented but are not reported here; the official SimMTM repository did not converge on this data regime.

5.3 Metrics

Forecasting is evaluated using RMSE, MAE, R^2 , and per-horizon/per-channel errors. The probabilistic head is evaluated with Gaussian NLL and empirical predictive coverage. Where target inferential

Table 2: Source-validation results. (a) Matched five-seed comparison of initializations (mean $\pm$ SD over seeds; lower is better; trivial baselines here and in Fig. 4a come from two separate evaluation scripts and differ slightly). (b) Selected ablations, single seed each on the pre-search base configuration; best tested setting per factor.

(a) Initialization (five seeds).		(b) Ablations (single seed).		
Model initialization	RMSE	Factor	Best tested setting	RMSE
Scratch	0.811 ± 0.022	Action injection	token	0.821
Pretrained body + fresh head	0.813 ± 0.022	Mask ratio	0.03	0.804
Complete pretrained checkpoint	0.812 ± 0.012	Latent weight	$\lambda_{\text{lat}} = 0.25$	0.774
Linear drift	0.928	VICReg	per-horizon, 0.1	0.798
Persistence	1.135	Transformer norm	pre-norm	0.829
		Horizon aggregation	pooled	0.787
		EMA	0.9995	0.799

statistics are reported, the 7 JOANNEUM runs are the independent units; windows are not used as independent replicates. R^2 is computed against the pooled mean of the evaluated channels and horizons.

6 Results

6.1 Corrected JEPA pretraining is stable but does not improve clean-source RMSE

A protocol audit identified three issues in the initial pretraining implementation: checkpoint selection based on training loss, latent schema consistency applied to an untrained physical head, and representational collapse in the EMA target encoder. The corrected pipeline selects checkpoints on deterministic held-out SSL validation, moves schema consistency to latent space, audits checkpoint loading, and applies VICReg to context and target representations. Target-encoder effective rank increased from approximately 5% to 58% of the latent dimensionality after correction.

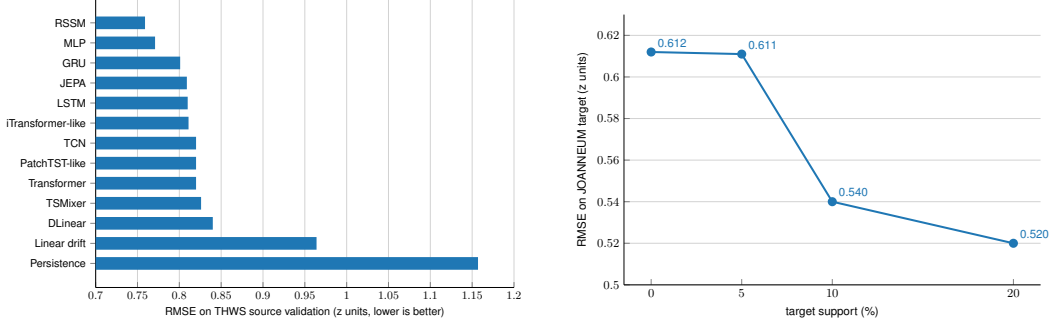
A matched five-seed comparison isolates the effect of pretraining (Table 2a). Training from scratch yields 0.811 ± 0.022 RMSE, pretrained body with a fresh head yields 0.813 ± 0.022 , and complete checkpoint transfer yields 0.812 ± 0.012 . The scratch-versus-pretraining difference is not significant ($p = 0.76$). Persistence and linear drift remain substantially weaker. Thus, no claim of improved clean-source forecasting from JEPA pretraining is supported by the current evidence.

6.2 Representation hyperparameters materially affect stability and forecasting

The latent-loss weight has a pronounced effect on downstream forecasting (Table 2b). The best tested value is $\lambda_{\text{lat}} = 0.25$ with RMSE 0.774; larger values degrade performance to 0.821 at $\lambda_{\text{lat}} = 1$ and 0.845 at $\lambda_{\text{lat}} = 4$. VICReg improves clean-source RMSE from 0.815 without regularization to 0.798 with per-horizon weighting and is additionally required to prevent collapse. Pooled horizon aggregation reaches 0.787 compared with 0.829 for per-horizon aggregation. Slow EMA targets are also preferable, with 0.9995 reaching 0.799 versus 0.808 at 0.999 and approximately 0.83 for faster updates. Two further observations temper the picture: removing the latent loss entirely ($\lambda_{\text{lat}} = 0$) yields 0.787, so the latent prediction objective at its default weight costs clean-source accuracy, and the best clean RMSE among mask ratios (0.804) is obtained with a very light mask (0.03), so the 0.35 mask of the search is retained for schema-drop robustness, not for clean accuracy. All ablations are single-seed runs on the pre-search base configuration. The locked candidate nevertheless keeps $\lambda_{\text{lat}} = 1$, per-horizon aggregation, mask ratio 0.35 and EMA 0.996, because the composite selection score (Sec. 5.2) weights schema-drop and masked robustness, calibration and action usage in addition to clean RMSE, and the discovery grid did not combine the ablation optima with schema consistency and action recovery. These ablations indicate that JEPA stability on CNC time series depends strongly on representation-objective balance rather than on a single default configuration.

6.3 Source-only model selection identifies a schema-consistent action-recovery variant as the locked configuration

The architecture search separates useful design choices from failure modes. The plain JEPA control, which removes anti-collapse regularization, collapses on 4/4 seeds and is excluded from ranking.



(a) Clean-source comparison on THWS validation. “-like” bars are in-house adaptations; the JEPA bar is the pre-search reference model on the source test split, all other bars are source-validation values.

(b) THWS→JOANNEUM few-shot adaptation of the pre-lock model. The 5% point is effectively zero-shot because the support subset holds no complete 48-step window.

Figure 4: Source forecasting benchmark (a) and pre-lock few-shot transfer (b).

Large-model scaling is also unstable in this data regime: the largest candidate ($d=512$, eight layers) collapses on 3/4 seeds. In contrast, the selected variant combines latent schema consistency and action recovery while preserving clean-source performance. Its source-validation score is 0.8216 ± 0.0089 RMSE with a schema-drop score among the strongest of the valid candidates (0.918) and clear action sensitivity (1.058 under action shuffling); its selection reflects the composite score rather than any single metric.

The stability rule is consequential. The per-horizon VICReg variant ranks first at the four-seed discovery stage, but the schema-consistency variant with action recovery overtakes it after five and then seven seeds. Locking on the initial discovery result would therefore have selected the wrong configuration. Figure 3 summarizes the sealed selection pipeline that keeps the JOANNEUM target data completely outside the architecture search.

6.4 Strong forecasting baselines remain competitive on the source machine

The clean-source forecasting benchmark is summarized in Fig. 4a. RSSM obtains RMSE 0.759 and MLP 0.771, outperforming the pre-search JEPA reference model at 0.809 (source test split) and the locked candidate at 0.822 (source validation). GRU, LSTM, and iTransformer-like models are also competitive. All learned models outperform linear drift and persistence. The result rules out a generic source-forecasting superiority claim and motivates evaluation on cross-machine adaptation and partial sensor overlap instead.

6.5 The locked model transfers zero-shot, but official zero-shot forecasters remain stronger in RMSE

The confirmatory JOANNEUM evaluation is executed once, after locking the schema + action-recovery configuration on source data only. The locked model reaches zero-shot RMSE 0.546, $R^2 = 0.012$, and NLL 0.52 on the target machine, improving over persistence at 0.654 (Table 3). This establishes that the transferred model beats a trivial baseline on an unseen machine despite the reduction from 17 to 10 available channels. The evaluated checkpoint is the single best-composite seed among the seven source-validation seeds, so the target number is a single-seed result. Per horizon, target R^2 is 0.035, 0.055, 0.052, -0.023 and -0.059 at $h = 1, 2, 4, 8, 16$: the transferred model explains a small share of variance at short horizons and falls below the pooled target-mean predictor at 8 and 16 s.

However, official PatchTST and iTransformer baselines with RevIN (Kim et al., 2022) achieve lower zero-shot RMSE, 0.503 and 0.498, respectively. The claim must therefore remain specific: the present advantage of SAAC-JEPA lies in action-conditioned latent dynamics, probabilistic outputs, and adaptation readiness rather than in state-of-the-art zero-shot RMSE.

Table 3: Confirmatory JOANNEUM zero-shot transfer after source-only model locking. Source column: official baselines are scored on the source test split (5,189 windows, 17 sensors); SAAC-JEPA is the seven-seed mean $\pm$ SD on source validation. Target column: 10 shared channels, 2,457 windows, one pass.

Method	Source RMSE	Target zero-shot RMSE	Notes
Persistence	—	0.654	trivial floor
PatchTST (official, RevIN)	0.804	0.503	deterministic
iTransformer (official, RevIN)	0.822	0.498	deterministic
SAAC-JEPA (locked)	0.822 ± 0.009	0.546	probabilistic, action-cond.

6.6 Few-shot adaptation remains promising but is still diagnostic

A pre-lock adaptation sweep suggests that limited target support can further improve transfer (Figure 4b). On the evaluated pre-lock model, zero-shot transfer yields RMSE 0.612, MAE 0.401, and $R^2 = -0.244$. With 10% target support, RMSE decreases to 0.540 and R^2 becomes positive (0.046). With 20% support, RMSE reaches 0.520, MAE 0.299, and $R^2 = 0.167$. The 5% support setting does not contain a valid 48-step support window and therefore reduces to the zero-shot model rather than constituting an adaptation failure.

7 Discussion

The evidence supports a narrow interpretation. First, JEPA pretraining does not improve clean-source forecasting over matched scratch training. Second, anti-collapse control and source-only selection stability are not minor implementation details: the plain JEPA control collapses completely, large scaling is unstable, and the provisional winner changes during confirmation. Third, cross-machine evaluation provides a distinct axis from in-domain forecasting. The locked schema + action-recovery model transfers zero-shot and outperforms persistence on an unseen machine, even though stronger zero-shot RMSE is currently achieved by official RevIN-equipped forecasting baselines.

Industrial predictive representations should therefore be judged on whether a learned state is controllable, uncertainty-aware, robust to schema shift, and adaptable under limited target data, not on source interpolation accuracy alone.

Future work. V2 should (i) train on more than one source machine so that invariance to machine-specific dynamics is measured rather than assumed; (ii) add instance-wise normalization or RevIN to SAAC-JEPA; (iii) connect the latent state to planning or model-based RL so it is judged by decision quality; and (iv) evaluate policies off-line before any real-machine test. Causal representation learning stays outside the claim set until the data provide enough intervention diversity (Fig. 5, App. F).

8 Limitations

One source and one target machine limit claims of general cross-machine generalization, and the 7 independent JOANNEUM runs restrict statistical power. The target stream is interpreted under a 2 ms controller-cycle assumption, and sensitivity to resampling remains a separate robustness analysis; seven source-only channels have no target counterpart. The few-shot curve was obtained before the model lock and must be re-measured on the locked configuration before being treated as confirmatory. The confirmatory target result is a single checkpoint evaluated once; run-level confidence intervals and action-sensitivity tests on the target machine have not yet been computed for the locked model. On the pre-lock model, zeroing or shuffling future actions changed target RMSE by less than 0.005, so action use under transfer is unverified and is the first item to test; that model was also under-calibrated (67% empirical coverage at nominal 90%). The ablations in Table 2b are single-seed, and the official baselines are single-seed runs without hyperparameter search. Finally, the present paper does not yet evaluate downstream policy learning or off-policy control, so the world-model interpretation remains predictive rather than decision-theoretic.

9 Conclusion

This study presents a schema-adaptive, action-conditioned JEPA with an audited cross-machine CNC protocol under partial sensor overlap. JEPA pretraining does not improve clean-source RMSE over matched scratch training; source-only search still identifies a stable schema-consistent, action-recovery configuration that transfers zero-shot to an unseen machine above persistence, while official RevIN-based forecasters remain stronger in raw zero-shot RMSE. The contribution is a controlled benchmark and methodology for industrial predictive representations under joint machine shift, schema shift, and action conditioning, not a state-of-the-art claim.

Code and data availability

Code, configuration files, the twenty candidate specifications, and the audit scripts are available at <https://github.com/ostertagmatthieu-dev/saac-jepa>; animated versions of the schematics are on the project page, <https://ostertagmatthieu-dev.github.io/saac-jepa/>. The source and target datasets are public under CC BY 4.0: THWS five-axis CNC milling (<https://doi.org/10.5281/zenodo.14094887>) and the FH JOANNEUM CNC machining repository (<https://doi.org/10.17632/gtvvwmz7r7.2>).

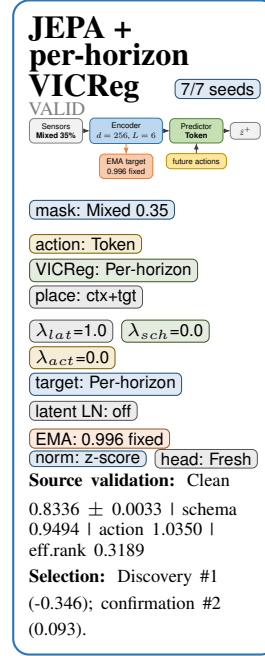
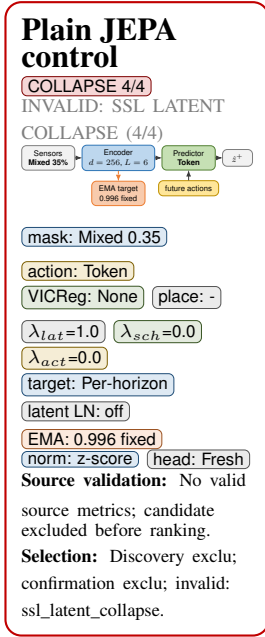
References

- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, pages 15619–15629, 2023.
- Adrien Bardes, Quentin Garrido, Jean Ponce, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *arXiv:2404.08471*, 2024.
- Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *ICLR*, 2022.
- Jiaxiang Dong, Haixu Wu, Haoran Zhang, Li Zhang, Jianmin Wang, and Mingsheng Long. SimMTM: A simple pre-training framework for masked time-series modeling. In *NeurIPS*, 2023.
- Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *ICLR*, 2023.
- Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. iTransformer: Inverted transformers are effective for time series forecasting. In *ICLR*, 2024.
- Taesung Kim, Jinhee Kim, Yunwon Tae, Cheonbok Park, Jang-Ho Choi, and Jaegul Choo. Reversible instance normalization for accurate time-series forecasting against distribution shift. In *ICLR*, 2022.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *ICML*, pages 2555–2565, 2019.
- Mario Martinez, Anna-Maria Schmitt, Andreas Schiffler, and Bastian Engelmann. Production data set for five-axis CNC milling with multiple changeovers. *Scientific Data*, 12:1067, 2025. doi:10.1038/s41597-025-05294-0. Dataset: <https://doi.org/10.5281/zenodo.14094887>.
- Markus Brillinger, Muaaz Abdul Hadi, Stefan Trabesinger, Johannes Schmid, and Florian Lackner. CNC machining data repository: Geometry, NC code & high-frequency energy consumption data for aluminum and plastic machining. *Data in Brief*, 61:111814, 2025. Dataset: <https://doi.org/10.17632/gtvvwmz7r7.2>.
- Peng Yan, Ahmed Abdulkadir, Gerrit A. Schatte, Giulia Aguzzi, Joonsu Gha, Nikola Pascher, Matthias Rosenthal, Yunlong Gao, Benjamin F. Grewe, and Thilo Stadelmann. Learning actionable world models for industrial process control. *arXiv:2503.01411*, 2025.

Appendix A. Architecture Search: Twenty Candidates

The architecture search contains 20 pre-defined SAAC-JEPA variants. Candidate ranking used **THWS source validation only**; the JOANNEUM target data were excluded from candidate ranking. The cards below show the common computational skeleton and highlight the mechanism changed by each candidate. Metrics are means over valid seeds from the architecture-search log. The final stability check reported in the closure log selected **the schema-consistency variant with action recovery** after seven valid seeds for the two leading candidates, after which code, configuration, normalizers, and checkpoint were frozen before the confirmatory target pass.

Reading the cards. Each diagram follows *masked sensor context* $\rightarrow$ *context encoder* $\rightarrow$ *action-conditioned predictor* $\rightarrow$ *future latent*, with an EMA target branch used for latent self-supervision. Tags report the masking regime, action-injection mechanism, latent objective, schema/action auxiliary losses, EMA schedule, capacity, normalization, and fine-tuning head initialization. Action sensitivity is $\text{RMSE}(\text{shuffled actions})/\text{RMSE}(\text{true actions})$; values above one indicate measurable action usage.



Schema-adaptive latent consistency

4/4 seeds

VALID



mask: Channel 0.35

action: Token

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.0$

target: Per-horizon

latent LN: off

EMA: 0.996 fixed

(norm: z-score) (head: Fresh)

Source validation: Clean

0.827 $\pm$ 0.0058 | schema

0.9252 | action 1.0437 |

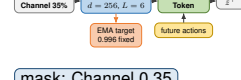
eff.rank 0.3792

Selection: Discovery #11
(-0.272).

Schema + action recovery

LOCKED

LOCKED WINNER



mask: Channel 0.35

action: Token

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.996 fixed

(norm: z-score) (head: Fresh)

Source validation: Clean

0.8216 $\pm$ 0.0089 | schema

0.9183 | action 1.0583 |

eff.rank 0.4273

Selection: Discovery #5
(-0.296); confirmation #1
(-0.830).

FiLM action injection

5/5 seeds

VALID



mask: Mixed 0.35

action: FiLM

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.996 fixed

(norm: z-score) (head: Fresh)

Source validation: Clean

0.8346 $\pm$ 0.0067 | schema

0.952 | action 1.0346 |

eff.rank 0.3289

Selection: Discovery #3
(-0.318); confirmation #4
(0.149).

Cross-attention action injection

5/5 seeds

VALID



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.996 fixed

(norm: z-score) (head: Fresh)

Source validation: Clean

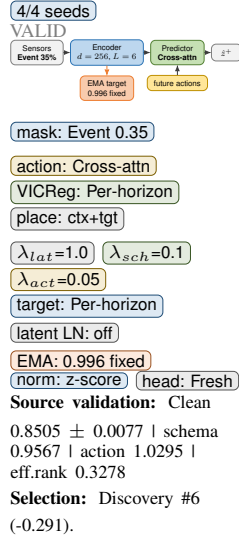
0.8333 $\pm$ 0.0034 | schema

0.9488 | action 1.0336 |

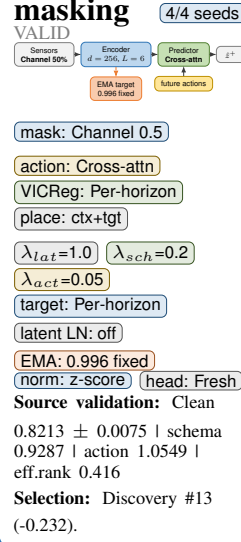
eff.rank 0.3149

Selection: Discovery #4
(-0.3); confirmation #3
(0.142).

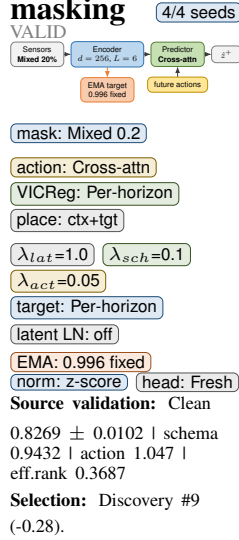
Event masking



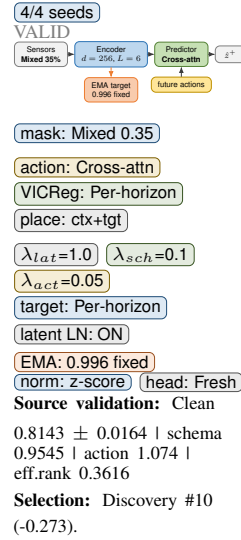
Heavy channel masking



Light mixed masking

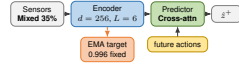


Latent LayerNorm ON



Pooled VICReg/target

4/4 seeds
VALID



mask: Mixed 0.35

action: Cross-attn

VICReg: Pooled

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$ target: Pooled

latent LN: off

EMA: 0.996 fixed

(norm: z-score) head: Fresh

Source validation: Clean

0.8139 $\pm$ 0.0117 | schema
0.9228 | action 1.079 |
eff.rank 0.4005

Selection: Discovery #14
(-0.229).

Low latent weight

4/4 seeds
VALID



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=0.5$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.996 fixed

(norm: z-score) head: Fresh

Source validation: Clean

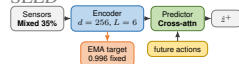
0.8019 $\pm$ 0.0088 | schema
0.9065 | action 1.0925 |
eff.rank 0.4145

Selection: Discovery #12
(-0.237).

High latent weight

1 COLLAPSE
PARTIAL: 1 COLLAPSED

SEED



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=2.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.996 fixed

(norm: z-score) head: Fresh

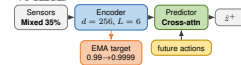
Source validation: Clean

0.8438 $\pm$ 0.0015 | schema
0.9499 | action 1.0214 |
eff.rank 0.2906

Selection: Discovery #2
(-0.323); confirmation #5
(1.177); invalid:
ssl_latent_collapse.

Scheduled EMA

4/4 seeds
VALID



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.99 $\rightarrow$ 0.9999

(norm: z-score) head: Fresh

Source validation: Clean

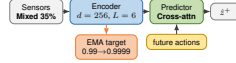
0.8345 $\pm$ 0.003 | schema
0.9574 | action 1.0311 |
eff.rank 0.3002

Selection: Discovery #7
(-0.291).

Robust source normalization

IQR PATHOLOGY
PATHOLOGICAL ROBUST

NORMALIZATION



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.99 → 0.9999

norm: Robust head: Fresh

Source validation: Clean

0.9813 $\pm$ 0.0360 | schema

0.9814 | action 1.0088 |

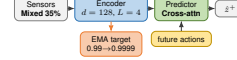
eff.rank 0.0213

Selection: Discovery #17
(6.25).

Compact model

4/4 seeds

VALID



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.99 → 0.9999

norm: z-score head: Fresh

Source validation: Clean

0.8485 $\pm$ 0.0006 | schema

0.9439 | action 1.0119 |

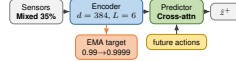
eff.rank 0.3485

Selection: Discovery #16
(0.462).

Medium-wide model

4/4 seeds

VALID



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.99 → 0.9999

norm: z-score head: Fresh

Source validation: Clean

0.8148 $\pm$ 0.0016 | schema

0.9325 | action 1.0758 |

eff.rank 0.305

Selection: Discovery #15
(-0.084).

Large model

COLLAPSE 3/4

UNSTABLE: SSL LATENT

COLLAPSE (3/4)



mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.99 → 0.9999

norm: z-score head: Fresh

Source validation: Clean

0.7937 (1 seed) | schema

0.8857 | action 1.1222 |

eff.rank 0.2462

Selection: Discovery #19
(-0.128); invalid:
ssl_latent_collapse.

Head-contamination control

4/4 seeds

VALID

Sensors Mixed 35%

Encoder $d = 256, L = 6$

Predictor Cross-attn

EMA target 0.99 $\rightarrow$ 0.9999

future actions

$\hat{z}^+$

mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.1$

$\lambda_{act}=0.05$

target: Per-horizon

latent LN: off

EMA: 0.99 $\rightarrow$ 0.9999

norm: z-score

head: Loaded

Source

validation: Clean 0.8359 $\pm$ 0.0036 | schema 0.9309 | action 1.0293 | eff.rank 0.2935

Selection: Discovery #8 (-0.288).

SAAC-JEPA full candidate

IQR PATHOLOGY

PATHOLOGICAL ROBUST

NORMALIZATION

Sensors Mixed 35%

Encoder $d = 384, L = 6$

Predictor Cross-attn

EMA target 0.99 $\rightarrow$ 0.9999

future actions

$\hat{z}^+$

mask: Mixed 0.35

action: Cross-attn

VICReg: Per-horizon

place: ctx+tgt

$\lambda_{lat}=1.0$ $\lambda_{sch}=0.2$

$\lambda_{act}=0.1$

target: Per-horizon

latent LN: off

EMA: 0.99 $\rightarrow$ 0.9999

norm: Robust

head: Fresh

Source validation: Clean

0.9557 $\pm$ 0.0013 | schema 0.9557 | action 1.0148 | eff.rank 0.011

Selection: Discovery #18 (6.3).

Appendix B. Complete Candidate Configuration Matrix

Table 4 lists the exact design factors varied across the twenty candidates. “ctx+tgt” denotes VICReg placement on context and EMA-target representations. All candidates use the same source-only data splits and evaluation protocol; only the listed architectural or training factors vary.

Table 4: Complete architecture-search configuration.

Candidate	Mask	Act.	VICReg	λ_{lat}	λ_{sch}	λ_{act}	EMA	d/L	Norm	LN	FT head
Plain control	mixed/0.35	Token	none	1.0	0.0	0.0	0.996	256/6	zscore	off	fresh
Per-horizon VICReg	mixed/0.35	Token	per-h	1.0	0.0	0.0	0.996	256/6	zscore	off	fresh
Schema consistency	channel/0.35	Token	per-h	1.0	0.1	0.0	0.996	256/6	zscore	off	fresh
Schema + action rec.	channel/0.35	Token	per-h	1.0	0.1	0.05	0.996	256/6	zscore	off	fresh
FiLM action	mixed/0.35	FiLM	per-h	1.0	0.1	0.05	0.996	256/6	zscore	off	fresh
Cross-attn action	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	0.996	256/6	zscore	off	fresh
Event masking	event/0.35	Cross-attn	per-h	1.0	0.1	0.05	0.996	256/6	zscore	off	fresh
Heavy channel mask	channel/0.5	Cross-attn	per-h	1.0	0.2	0.05	0.996	256/6	zscore	off	fresh
Light mixed mask	mixed/0.2	Cross-attn	per-h	1.0	0.1	0.05	0.996	256/6	zscore	off	fresh
Latent LayerNorm	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	0.996	256/6	zscore	on	fresh
Pooled VICReg	mixed/0.35	Cross-attn	pool	1.0	0.1	0.05	0.996	256/6	zscore	off	fresh
Low latent weight	mixed/0.35	Cross-attn	per-h	0.5	0.1	0.05	0.996	256/6	zscore	off	fresh
High latent weight	mixed/0.35	Cross-attn	per-h	2.0	0.1	0.05	0.996	256/6	zscore	off	fresh
Scheduled EMA	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	sched.	256/6	zscore	off	fresh
Robust normalization	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	sched.	256/6	robust	off	fresh
Compact model	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	sched.	128/4	zscore	off	fresh
Medium-wide model	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	sched.	384/6	zscore	off	fresh
Large model	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	sched.	512/8	zscore	off	fresh
Loaded-head control	mixed/0.35	Cross-attn	per-h	1.0	0.1	0.05	sched.	256/6	zscore	off	loaded
Full candidate	mixed/0.35	Cross-attn	per-h	1.0	0.2	0.1	sched.	384/6	robust	off	fresh

Appendix C. Source-Validation Search Outcomes

The table below reports the source-only architecture-search outcomes. Composite scores are z-normalized within their respective selection pools and therefore should not be compared across discovery and confirmation columns. The plain control was excluded because all four seeds collapsed. The large model collapsed in three of four discovery seeds. The two robust-normalization candidates exhibited the pathology associated with near-zero IQR channels. The final closure-stage stability check selected the schema-consistency variant with action recovery after seven valid seeds for the leading pair.

Table 5: Source-validation outcomes for the twenty candidates. Lower RMSE/schema-drop is better; action sensitivity greater than one indicates that shuffled future actions degrade prediction.

Candidate	Seeds	Clean RMSE	Schema	Masked	Long	Calib.	Action sens.	Eff. rank	Discovery / confirm.
Plain control	0/4	—	—	—	—	—	—	—	excluded / excluded
Per-horizon VICReg	7/7	0.8336 $\pm$ 0.0033	0.9494	0.8411	0.8797	0.1035	1.0350	0.3189	#1 (-0.346) / #2 (0.093)
Schema consistency	4/4	0.827 $\pm$ 0.0058	0.9252	0.8344	0.8747	0.1245	1.0437	0.3792	#11 (-0.272) / —
Schema + action rec.	7/7	0.8216 $\pm$ 0.0089	0.9183	0.8298	0.8709	0.1315	1.0583	0.4273	#5 (-0.296) / #1 (-0.830)
FiLM action	5/5	0.8346 $\pm$ 0.0067	0.952	0.8412	0.8793	0.0957	1.0346	0.3289	#3 (-0.318) / #4 (0.149)
Cross-attn action	5/5	0.8333 $\pm$ 0.0034	0.9488	0.8415	0.8786	0.1003	1.0336	0.3149	#4 (-0.3) / #3 (0.142)
Event masking	4/4	0.8505 $\pm$ 0.0077	0.9567	0.8492	0.8976	0.125	1.0295	0.3278	#6 (-0.291) / —
Heavy channel mask	4/4	0.8213 $\pm$ 0.0075	0.9287	0.8297	0.8713	0.1407	1.0549	0.416	#13 (-0.232) / —
Light mixed mask	4/4	0.8269 $\pm$ 0.0102	0.9432	0.8364	0.8752	0.1171	1.047	0.3687	#9 (-0.28) / —
Latent LayerNorm	4/4	0.8143 $\pm$ 0.0164	0.9545	0.8216	0.866	0.1268	1.074	0.3616	#10 (-0.273) / —
Pooled VICReg	4/4	0.8139 $\pm$ 0.0117	0.9228	0.8221	0.8657	0.1435	1.079	0.4005	#14 (-0.229) / —
Low latent weight	4/4	0.8019 $\pm$ 0.0088	0.9065	0.8127	0.8535	0.1421	1.0925	0.4145	#12 (-0.237) / —
High latent weight	4/5	0.8438 $\pm$ 0.0015	0.9499	0.8521	0.8865	0.0976	1.0214	0.2906	#2 (-0.323) / #5 (1.177)
Scheduled EMA	4/4	0.8345 $\pm$ 0.003	0.9574	0.842	0.8804	0.0807	1.0311	0.3002	#7 (-0.291) / —
Robust normalization	4/4	0.9813 $\pm$ 0.0360	0.9814	0.9816	1.0126	0.0399	1.0088	0.0213	#17 (6.25) / —
Compact model	4/4	0.8485 $\pm$ 0.0006	0.9439	0.856	0.8883	0.0666	1.0119	0.3485	#16 (0.462) / —
Medium-wide model	4/4	0.8148 $\pm$ 0.0016	0.9325	0.8237	0.8686	0.1147	1.0758	0.305	#15 (-0.084) / —
Large model	1/4	0.7937 (1 seed)	0.8857	0.8018	0.8558	0.1156	1.1222	0.2462	#19 (-0.128) / —
Loaded-head control	4/4	0.8359 $\pm$ 0.0036	0.9309	0.8416	0.8818	0.0818	1.0293	0.2935	#8 (-0.288) / —
Full candidate	4/4	0.9557 $\pm$ 0.0013	0.9557	0.9558	0.9866	0.0075	1.0148	0.011	#18 (6.3) / —

Appendix D. Architecture-Search Interpretation

Anti-collapse is necessary. The plain control removes VICReg and collapses for all four discovery seeds. This matches the corrected pretraining audit, in which target-encoder effective rank increased from approximately 5% to 58% after anti-collapse regularization was moved to the appropriate latent representations.

Capacity scaling is not monotonic. The large model ($d = 512$, eight layers) collapses in three of four seeds, whereas the locked candidate uses $d = 256$ and six layers. The search therefore does not support a “larger is better” interpretation at the available source-data scale.

Robust normalization requires channel-specific safeguards. The two robust-normalization candidates produce pathological errors because near-constant source channels can have an interquartile range close to zero. These candidates are retained in the appendix because they document a real failure mode of the search space rather than an omitted negative result.

The locked candidate reflects the intended inductive bias. The locked candidate combines channel masking, latent schema consistency, explicit action recovery, per-horizon VICReg on context and target representations, and a fresh downstream head. Its source-validation advantage is not defined by the lowest clean RMSE alone; the selection score also rewards schema-drop robustness, masked robustness, long-horizon performance, calibration, SSL validation, and action use.

Appendix E. Official Forecasting Baseline Protocol

The official PatchTST (yuqinie98/PatchTST, commit 204c21e, supervised tree) and iTransformer (thuml/iTransformer, commit c2426e6) repositories were run through a shared dataset adapter so that both consume the same session-bounded windows as SAAC-JEPA: context 32 steps, prediction length 16, scored at horizons $\{1, 2, 4, 8, 16\}$, 1 Hz grid, windows never crossing a session boundary (18,267 / 3,423 / 5,189 source windows and 2,457 target windows, identical to the SAAC-JEPA counts). Inputs are the 17 canonical sensors plus the 4 action channels (21 channels, all forecast jointly); RMSE is reported on the 17 sensors for the source test split and on the 10 channels shared with the target for the sealed target pass. Values are z-scored with the source-train normalizer of the main pipeline; the 7 sensors absent on the target are zero-imputed in normalized space and excluded from scoring. Both baselines were trained once (single seed 20260820, 100 epochs, early-stopping patience 15, batch 128, MSE loss, RevIN enabled without affine parameters, other hyperparameters at repository defaults) with no hyperparameter search. The official SimMTM repository (commit 169513b) diverged to NaN at two learning rates under the same protocol and is therefore not reported. Persistence on the same target windows gives RMSE 0.654; on the source test split it gives 1.128 (17 sensors). The official baselines forecast the action channels poorly (target RMSE 2.01 and 2.27), which is expected because commanded actions are exogenous inputs rather than process outcomes.

Appendix F. Concept of Study and V2 Roadmap

The overall concept of the study is a single pipeline from historical machine data to a deployable control decision:

1. **Source machine:** real production data (sensors and commanded actions) from the source CNC machine.
2. **SAAC-JEPA:** schema-adaptive, action-conditioned JEPA pretraining on the source data.
3. **Compact representation of the dynamics:** a latent state that summarizes the machine’s physical evolution.
4. **Transfer to the target machine:** zero-shot use of the source-trained representation under partial sensor overlap.
5. **Adaptation with a few target samples:** few-shot or online updates using limited data from the new machine.
6. **World model:** the adapted latent dynamics used as a simulator of the target machine.
7. **Simulation of candidate actions:** rolling out alternative command sequences in the world model.
8. **Reinforcement learning:** policy optimization inside the learned simulator.
9. **New control policy:** the resulting candidate policy for the target machine.
10. **Doubly robust evaluation on historical data:** off-policy evaluation of the candidate policy with historical logs before any physical test.
11. **Decision policy:** the evaluated policy that is retained for deployment or decision support.

The present V1 paper covers steps 1–5 (steps 4 and 5 as zero-shot and pre-lock few-shot evidence). Figure 5 shows how V2 extends this backbone.

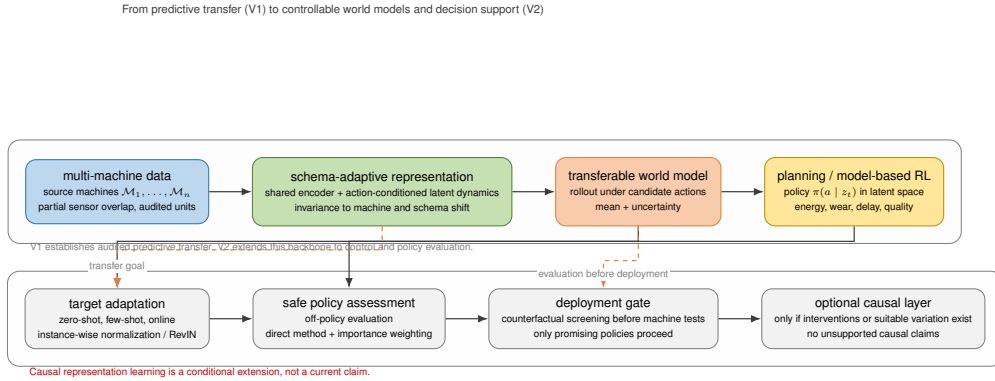


Figure 5: V1 establishes audited cross-machine predictive transfer with schema-adaptive latent dynamics. V2 extends this backbone to multi-machine representation learning, target adaptation with stronger normalization, action-conditioned planning or model-based RL, and policy assessment through off-policy evaluation before any physical deployment. Optional causal representation learning is shown as a conditional branch rather than as a current contribution.